



MARTIN-LUTHER-UNIVERSITÄT
HALLE-WITTENBERG

Dr. Juliane Hennecke

Chair of Empirical Microeconomics

ADVANCED MICROECONOMICS

Decisions under Risk and Uncertainty II - Risk Preferences

Winter Term 2025/2026

Outline

1) Consumer Theory

- 1.1) Preferences and Utility (L1)
- 1.2) Utility Maximization (L2)
- 1.3) Expenditure Minimization (L3)
- 1.4) Demand Functions and Duality (L4+L5)

2) General Equilibrium

- 2.1) Exchange Economy (L6)
- 2.2) Welfare Economics and General Equilibrium (L7)

3) Decisions under Risk and Uncertainty

- 3.1) Expected Utility (L8)
- 3.2) Risk Preferences (L9)
- 3.3) Insurances (L10)
- 3.4) Safety Regulations and the Value of a Statistical Life (11)

4) Behavioral Economics (L12)

Literature

- Autor, David (2016): Lecture Notes 17
- Anger, Erik (2021): A Course in Behavioral Economics, 3rd edition, Palgrave MacMillan, Chapter 4 and 6
- **Dohmen et al. (2011)**: Individual risk attitudes: Measurement, determinants, and behavioral consequences. JoEEA, 9(3): 522–550.
- Binswanger (1981) "Attitudes toward Risk: theoretical implications of an experiment in rural India." Economic Journal 91, 867–890.
- Fezzi et al. (2021) "Estimating risk preferences integrating insurance choices with subjective beliefs" European Economic Review, 135: 103717
- Holt and Laury (2002) "Risk aversion and incentive effects." American Economic Review 92, 1644–1655.
- Holt and Laury (2014) "Assessment and Estimation of Risk Preferences" Handbook of Risk and Uncertainty, Chapter 4
- Kachelmeier, Shehata (1992) "Examining Risk Preferences Under High Monetary Incentives: Experimental Evidence from the People's Republic of China" The American Economic Review 82(5): 1120–1141
- Kahneman, D., Tversky, A. (1979) "Prospect theory: an analysis of decision under risk." Econometrica 47, 263–291.

Recap: Expected utility theory

- **Expected utility theory** allows modeling decisions over risky bundles of goods.
- This theory relies on **strong assumptions**
 - a finite number of possible states of the world that occur with known probabilities.
 - Consequentialism: Only the reduced lotteries are relevant.
 - Consumers have rational and continuous preferences over reduced lotteries that also fulfill the independence axiom.
- Under these assumptions expected utility theory results in **intuitive utility functions** that **only depend on final outcomes**

Motivation

Imagine I offered you the following game: We **toss a fair coin**.

- If head comes up, I'll pay you €1 million.
- If tail comes up you'll have to pay me €0.8 million.

Would you like to participate in this game?



Quelle: Wikipedia

Why does theory discussed thus far not suffice?

- **So far:** Decision over bundles of goods (=alternatives).
- Thus we needed preferences over these bundles.
- **Now:** Decision over **risky** bundles of goods.
- In addition to preferences over bundles of goods, we need preferences over risk: **risk preferences**.
- **Expected utility theory** models decision under risk as an extension of the theory that we already know.
- It builds on known utility functions and allows to incorporate risk – under certain additional assumptions on preferences.

Overview

1 Risk Preferences

2 Measuring Risk Preferences

Experiments

Observational data

Survey Question

3 Application: Dohmen et al. 2011

Risk Preferences

- The shape of the utility function describes the **preferences of individuals**.
- In decisions under risk, choices are determined by the **curvature of the utility function**.
- The utility function contains the individual's **risk preference**.
- Preferences can be **risk-averse, risk-loving, or risk-neutral**.
- Since different individuals may have different utility functions, it is rational that people facing the same situation make different choices — as long as they maximize their expected utility.

Risk Preferences

Risk Preferences

Individual feelings towards risk which enter expected utility theory through the utility function.

Suppose a lottery and a certain (sure) alternative have the same expected value. Then an individual is ...

- **risk-averse**: if they prefer the certain alternative over the lottery
- **risk-loving**: if they prefer the lottery over the certain alternative
- **risk-neutral**: if they are indifferent between the certain alternative and the lottery

Risk Preferences

- Risk preferences (w.r.t. a specific lottery) are reflected in the

Certainty equivalent

c of a lottery $L = (p_1, p_2)$ with outcomes x_1 and x_2 is the amount of money that - when received with certainty - gives an individual the same expected utility as the lottery. c is implicitly defined by

$$U(c) = p_1 U(x_1) + p_2 U(x_2).$$

Risk premium

r is the amount of money an individual is willing to give up to switch from a situation with risk (the specific lottery) to a situation without risk. It thus is the difference between the expected value of the lottery and the certainty equivalent:

$$r = p_1 x_1 + p_2 x_2 - c.$$

Risk preferences

Examples

- Take the following utility functions:
 - ① $U(w) = w^{0.5}$
 - ② $U(w) = w$
 - ③ $U(w) = w^2$
- Decision makers with the different utility functions are offered a gamble (a lottery) that pays them €2 with probability 50%, and 0 otherwise.
 - Expected value of the lottery: $0.5 \times 2 + 0.5 \times 0 = 1$
- What is the certainty equivalent of this lottery for the different decision makers? And what is the risk premium?

Risk preferences

$$U(w) = w^{0.5}$$

- Expected utility of the lottery:
 $0.5 \times u(2) + 0.5 \times u(0) = 0.5 \times \sqrt{2} + 0.5 \times 0 = 0.707$
- We are looking for **certainty equivalent** c , with $U(c) = 0.5 \times \sqrt{2}$:

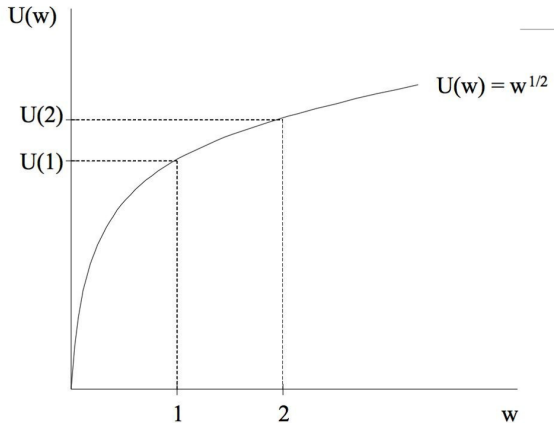
$$c^{0.5} = 0.5 \times \sqrt{2}$$

$$c = (0.5 \times \sqrt{2})^2 = 0.5$$

- Here $c < \text{expected value}$, and the **risk premium** is
 $r = 1 - 0.5 = 0.5$.
- The decision maker is **risk-averse**
 - The decision maker with this utility function gets a disutility from risk (would be willing to pay 0.50 to not play the game)

Risk preferences

$U(w) = w^{0.5}$ - concave utility function



Risk preferences

$$U(w) = w$$

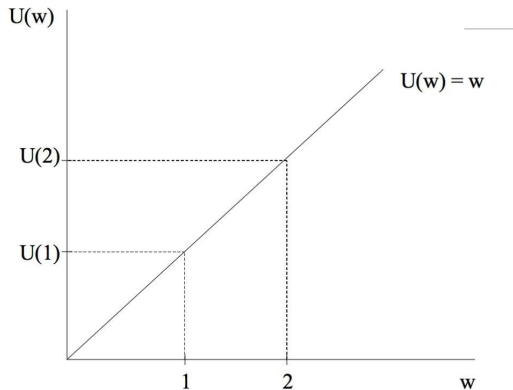
- Expected utility of the lottery:
 $0.5 \times u(2) + 0.5 \times u(0) = 0.5 \times 2 + 0.5 \times 0 = 1$
- We are looking for **certainty equivalent** c , with $U(c) = 1$:

$$c = 1$$

- Here $c =$ expected value, and the **risk premium** is $1 - 1 = 0$.
- The decision maker is **risk-neutral**
 - The decision maker with this utility function gets neither utility nor disutility from risk (would be willing to pay nothing to (not) play the game)

Risk preferences

$$U(w) = w$$



Risk preferences

$U(w) = w^2$ - **convex utility function**

- Expected utility of the lottery:
 $0.5 \times u(2) + 0.5 \times u(0) = 0.5 \times 2^2 + 0.5 \times 0 = 2$
- We are looking for **certainty equivalent** c , with $U(c) = 0.5 \times 2^2$:

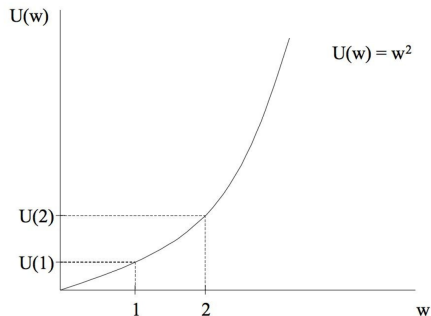
$$c^2 = 2$$

$$c = \sqrt{2} = 1.414$$

- Here $c > \text{expected value}$, and the risk premium is
 $1 - 1.414 = -0.414$.
- The decision maker is **risk-loving**.
 - DM with this utility function gets a positive utility from risk (would be willing to pay 0.414 to play the game).

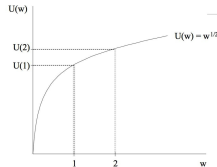
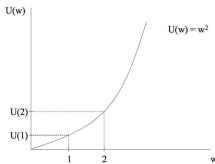
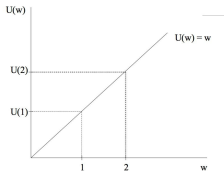
Risk preferences

$$U(w) = w^2$$



Risk preferences - Utility Functions

- The **shape of the utility function** (concave, convex, linear) captures the decision maker's risk preferences:
 - **Risk neutral:** linear function
 - **Risk loving:** convex function
 - **Risk averse:** concave function



Risk preferences - Jensen's Inequality

Jensen's Inequality

Risk preferences can be summarized as

- Risk neutral: $E(U(w)) = U(E(w))$
- Risk loving: $E(U(w)) > U(E(w))$
- Risk averse: $E(U(w)) < U(E(w))$

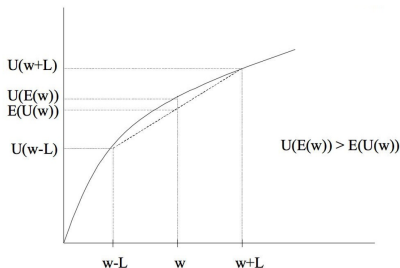
where

$E(U(w)) = \sum_{i=1}^N p_i U(w_i)$ (expected utility of the lottery)

$U(E(w)) = U\left(\sum_{i=1}^N p_i w_i\right)$ (utility of expected value of the lottery).

Risk preferences - Graphical illustration

- **Risk-averse decision maker** - Concave utility function (diminishing marginal utility)



- Loss of L compared to w is worse than winning L .
- DM prefers the expected value of the lottery with certainty over the lottery itself.
- Consumer is better off receiving a given amount of wealth with certainty than the same amount of wealth on average with a variance around it.

Implications & What Comes Next

- We have developed substantial analytical machinery to model choices under risk — for a seemingly simple goal.
- This machinery becomes powerful in the lectures to come:
 - Using VNM Expected Utility, we can formally derive individuals' **willingness to pay to avoid, reduce, or transfer risk**.
 - This allows us to analyze
 - **explicit markets for risk**, such as insurance markets [*Lecture 10*] and
 - **implicit markets for risk** that help us learn about the (statistical) value of life [*Lecture 11*]

Overview

1 Risk Preferences

2 Measuring Risk Preferences

Experiments

Observational data

Survey Question

3 Application: Dohmen et al. 2011

Overview

1 Risk Preferences

2 Measuring Risk Preferences

Experiments

Observational data

Survey Question

3 Application: Dohmen et al. 2011

Measuring Risk Preferences in Experiments

- **Revealed Preferences** - Individuals **reveal** their risk preferences by choices between safe and risky options
- **Measurement** can be implemented in different ways (see Holt and Laury, 2014)
 - Binary choices
 - Investment tasks
 - Structured binary choices (probability choice menu)
 - Price-based assessment of certainty equivalent (pricing task)
- Choices are typically **incentivized** (*people get paid for participation and pay depends on choices*)

Experiments (1) - Binary choice experiment

- Experiments with **simple binary choice gambles** (Kahneman & Tversky (1979)), i.e.
- Do you want
 - A** \$3 for sure
 - B** 0.8 chance of \$4 and \$0 otherwise
- Expected values: $EV(A) = 3$, $EV(B) = 0.8 \times 4 = 3.2$
- **General idea:** The gambles are constructed to have similar expected values but different level of risk.
- Choices therefore reveal each participant's tolerance for risk based on how much variability they are willing to accept.
- What does **choice of safe option** reveal?
 - Risk aversion (*in the gain domain*)
 - A concave utility function: $U(3) > 0.8U(4) + 0.2U(0)$.
 - People “dislike” variability in gains and prefer a sure smaller gain over a risky larger gain.

Experiments (2) - Investment task

- Participants make **investment decision** by deciding between different gambles
- Choose one of the following gambles:

	with 50-50
Gamble 0	50
Gamble 1	45 or 95
Gamble 2	40 or 120
Gamble 3	30 or 150
Gamble 4	10 or 190
Gamble 5	0 or 200

- Gambles move from conservative (Gamble 0) to very risky (Gamble 5) as the downside risk increases.
- **Empirical Evidence:** Binswanger (1981)
 - Experiments with low-income farmers in Bangladesh
 - **Results:** Among the participants 80% chose Gamble 2 or 3 (moderately risk averse) and nobody chose 5

Experiments (3) - Structured binary choices

- Participants make a sequence of paired choices where the probability of the high payoff increases across rows.
- This allows inference of risk preferences based on the **switching point** from the safer Option A to the riskier Option B.
- Make a choice for each situation

	Option A	Option B
1	1/10 \$2 and 9/10 \$1.60	1/10 \$3.85 and 9/10 \$0.10
2	2/10 \$2 and 8/10 \$1.60	2/10 \$3.85 and 8/10 \$0.10
3	3/10 \$2 and 7/10 \$1.60	3/10 \$3.85 and 7/10 \$0.10
	...	
9	9/10 \$2 and 1/10 \$1.60	9/10 \$3.85 and 1/10 \$0.10
10	10/10 \$2 and 0/10 \$1.60	10/10 \$3.85 and 0/10 \$0.10

- **Empirical Evidence:** Holt and Laury (2002)
 - **Results:** Participants generally switch rather late → consistent with **risk aversion**.

Experiments (4) - Price-based assessment of certainty equivalent

- Individuals are given a gamble
- They are asked which amount of money they want, to give the gamble back to experimenter (certainty equivalent)
- Then a random amount is drawn
 - If larger than or equal to an individual's stated amount, individual gets the amount as payout
 - Otherwise the gamble determines the individual's payout
 - **Empirical Evidence:** Kachelmeier and Shehata (1992) - Experiment run in China
 - Result: Individuals were risk loving (stated selling prices higher than expected value of gamble)
 - Possible interpretation: "Endowment effect"

Experiments (5) - Multiple-price list to elicit certainty equivalent

- Participants make a choice in each row between a 50–50 lottery and a certain payment.
- The certain amount increases across rows → reveals the **certainty equivalent** and thus risk attitudes.
- Make a choice for each situation:

	Option A (50-50)	Option B
1	€300 or 0	€0
2	€300 or 0	€10
3	€300 or 0	€20
...		
19	€300 or 0	€180
20	€300 or 0	€190

- A risk-neutral individual is indifferent when $EV(A) = EV(B)$, i.e. expected value of gamble equals expected value of certain outcome
→ They switch at $B = 150$, risk-averse individuals should switch earlier
- **Empirical Evidence:** Dohmen et al. (2011)

Overview

1 Risk Preferences

2 Measuring Risk Preferences

Experiments

Observational data

Survey Question

3 Application: Dohmen et al. 2011

Observational data

- Risk preferences can also be inferred from **real-life choices** based on observational data
- Choices that are often used in this context: insurance choices
- **Example:** Fezzi et al. (2021) use decisions of crop-insurance (against bad weather events) of Italian apple farmers
 - Farmers have three options
 - ① no insurance (NI)
 - ② lower-premium insurance only against damages from hail (IH)
 - ③ higher-premium insurance against damages from hail and frost (IT)

Observational data - Fezzi et al. (2021)

- **Structural approach** - Researchers explicitly specify the underlying decision model and then estimate the unknown parameters of that model using observed choices.
- Decision **theoretically modeled** using expected utility theory: Individuals choose insurance option with highest expected utility (which depends on profits π that in turn depend on damages from hail and frost)

$$EU(\pi) = \int \int u(\pi) f_1(x_{hail}) f_2(x_{frost}) dx_{hail} dx_{frost}$$

- Survey on farmers beliefs about f_1 and f_2 , assumption for $u()$ and then use farmer's insurance choices to estimate parameters of $u()$, including risk aversion
- **Results:** Farmers are on average moderately risk averse

Overview

1 Risk Preferences

2 Measuring Risk Preferences

Experiments

Observational data

Survey Question

3 Application: Dohmen et al. 2011

Measuring risk preferences using survey question

- **“How much risk are you willing to take in general?”**
 - Rate on a scale from 0 (*not willing to take risks at all*) to 10 (*very willing to take risks*)
- Widely used, e.g. in the *German Socio-Economic Panel (SOEP)* and many other international household panel studies
- **Advantage:** Simple, fast, and low-cost measurement
- **Disadvantage:** Not incentivized → concerns about validity
- **Empirical Evidence:** Dohmen et al. (2011)

Overview

1 Risk Preferences

2 Measuring Risk Preferences

Experiments

Observational data

Survey Question

3 Application: Dohmen et al. 2011

Dohmen et al. (2011), JEEA, 9(3):522–550

INDIVIDUAL RISK ATTITUDES: MEASUREMENT, DETERMINANTS, AND BEHAVIORAL CONSEQUENCES

Thomas Dohmen
Maastricht University

David Huffman
Swarthmore College

Jürgen Schupp
DIW, Berlin,
and Free University of Berlin

Armin Falk
University of Bonn

Uwe Sunde
University of St. Gallen

Gert G. Wagner
Max Planck Institute for Human
Development, Berlin

Abstract

This paper studies risk attitudes using a large representative survey and a complementary experiment conducted with a representative subject pool in subjects' homes. Using a question asking people about their willingness to take risks "in general", we find that gender, age, height, and parental background have an economically significant impact on willingness to take risks. The experiment confirms the behavioral validity of this measure, using paid lottery choices. Turning to other questions about risk attitudes in specific contexts, we find similar results on the determinants of risk attitudes, and also shed light on the deeper question of stability of risk attitudes across contexts. We conduct a horse race of the ability of different measures to explain risky behaviors such as holdings stocks, occupational choice, and smoking. The question about risk taking in general generates the best all-round predictor of risk behavior. (JEL: D0, D81, C90, J10)

Dohmen et al. (2011)

- We will discuss the following questions:
 - What is the paper about?
 - How is it related to our class?
 - What data and methods are used?
 - What are the paper's (main) results?
 - How valid are the paper's findings?
 - What is your personal take-away from this paper?

1. What is the paper about?

2. How is it related to our class?

3. What data and methods are used?

4. What are the paper's (main) results?

5. How valid are the paper's findings?

6. What is your personal take-away from this paper?

Summary

- Risk preferences determine choices under uncertainty.
- Expected Utility Theory links choices to the curvature of the utility function.
- Certainty equivalents and risk premia provide intuitive measures of risk attitudes.
- Risk preferences can be measured in three main ways:
 - Experiments (incentivized choices)
 - Observational data (real-world choices, i.e. insurance choices)
 - Survey questions (e.g. SOEP general risk question)
- Each method has advantages and limitations regarding ease of implementation and validity.
- Empirical evidence: Most individuals are moderately risk-averse, but attitudes vary across contexts, stakes, and domains.

Important Keywords

- Expected Utility Theory
- Risk Preferences
- Risk Aversion / Neutrality / Loving
- Utility Function Curvature (concave, linear, convex)
- Certainty Equivalent
- Risk Premium
- Jensen's Inequality
- Revealed Preferences
- Incentivized Experiments
- Multiple Price List (MPL)
- Structural Estimation
- Subjective Beliefs
- Survey-Based Risk Measures
- Validity & Predictive Power

Questions?

